
Welcome to Calculus I

Work these in class. Worked solutions are on the posted notes page — each problem below names the section heading to look for.

Day 1 • Problem 1*see: Big Limit #1: How Fast, Right Now?*

Over an *interval* you can do it with algebra you already have. Using $s(t) = 16(-t^2 + 2t + 3)$, compute the average velocity

$$\frac{s(x) - s(a)}{x - a}$$

with $a = 0.5$ — first for $x = 1$, then for $x = 0.6$.

Day 1 • Problem 2*see: Do It Once with Letters*

Form $\frac{s(x) - s(0.5)}{x - 0.5}$ using $s(t) = 16(-t^2 + 2t + 3)$, and simplify it as far as it will go. Factor the top.

Day 1 • Problem 3 (worked together)*see: One Triangle at a Time: the Square*

I will do the square completely. Watch the *method*, because you are about to use it again on the hexagon.

Day 1 • Problem 4*see: Your Turn: the Hexagon*

Step 1. The six radii cut the hexagon into six triangles. Take one: how long are its three sides, and so what kind of triangle is it?

Day 1 • Problem 5*see: Your Turn: the Hexagon*

Step 2. Find the area of that one triangle. Its base is 1; use the 30–60–90 triangle from earlier to get its height.

Day 1 • Problem 6

see: Your Turn: the Hexagon

Step 3. Finish it — what is the area of the whole hexagon?

Day 1 • Problem 7

see: Your Turn: the Hexagon

The circle's area is $\pi \approx 3.14159$. How do your two answers compare with it, and which is closer?

Day 1 • Problem 8 (worked together)

see: Whose Example Is It?

I work it at the board — usually because it introduces a new idea, or because watching once beats being stuck for ten minutes.

Day 1 • Problem 9

see: The Experiment You Are About to Run on Yourself

Predict: on that exam, how did the unrestricted-AI group do against the group that had no AI at all?